

Switch IP atama

```
configure terminal  
interface vlan 99  
ip address IP_ADRESI AG_MASKESI  
no shutdown,
```

Switch ağ geçidi ayarlama:

```
configure terminal  
ip default-gateway IP_ADRESI
```

Router arayüz yapılandırma

```
interface gigabitEthernet 0/0/0  
ip address IP_ADRESI AG_MASKESI  
description ACIKLAMA_MESAJI  
no shutdown
```

Ayrıcalıklı mod şifreleme:

```
enable password SIFRE  
enable secret SIFRE
```

Konsol Kablosu şifreleme:

```
line console 0  
password SIFRE  
login
```

Telnet hattını şifreleme:

```
line vty 0 15 (Router'da 0 4)  
password SIFRE  
login
```

Switch'te SSH desteği var mı? show ip ssh

SSH yapılandırma

```
ip domain-name cisco.com  
crypto key generate rsa → 1024  
username K_ADI secret SIFRE  
line vty 0 15  
transport input ssh → Telneti devre dışı bırakır.
```

```
login local  
exit  
ip ssh version 2
```

Switch VLAN yapılandırma

```
configure terminal  
vlan ID  
name NAME  
end
```

Switch VLAN port atama

```
configure terminal  
interface ID  
switchport mode access  
switchport access vlan ID  
end
```

**** Ses trafiği yapılandırma:**

```
configure terminal  
interface ID  
switchport mode access  
mls qos trust cos  
switchport voice vlan ID  
end
```

**** Switch'teki bir portun VLAN'ını sıfırlama:**

```
interface fa0/18  
no switchport access vlan  
end
```

**** Switch MDIX ayarı:**

```
configure terminal  
interface ID  
duplex full  
speed 100  
mdix auto
```

VLAN Trunk yapılandırma:

```
configure terminal  
interface ID  
switchport mode trunk
```

switchport trunk native vlan ID → Native VLAN'ı ID yapar.

switchport trunk allowed vlan IDLER → Trunk bağlantısı için izin verilecek VLAN IDleri.

Trunk sıfırlama:

```
interface fa0/1
```

```
no switchport trunk allowed vlan
```

```
no switchport trunk native vlan
```

Switch DTP ayarları:

```
configure terminal
```

```
interface fa0/1
```

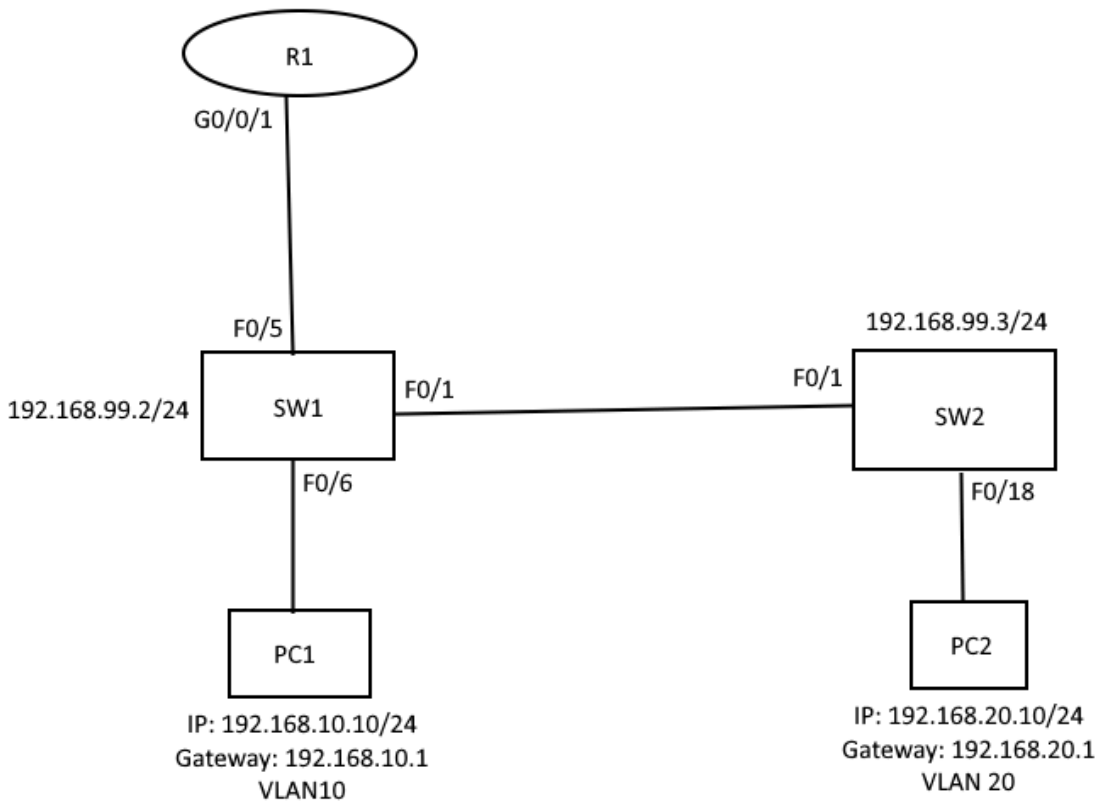
```
switchport mode trunk
```

```
switchport mode dynamic desirable → DTP aktif olarak aktif etkinleştir.
```

```
switchport mode dynamic auto → DTP pasif olarak aktif. Her cihaza yanıt verir.
```

```
switchport nonegotiate → DTP kapatır.
```

VLAN'ler arası yönlendirme:



SW1 yapılandırma:

vlan 10

name LAN10

vlan 20

name LAN20

vlan 99

name Management

interface vlan 99

ip address 192.168.99.2 255.255.255.0

no shutdown

exit

ip default-gateway 192.168.99.1

interface fa0/6

switchport mode access

switchport access vlan 10

no shutdown

interface fa0/1

switchport mode trunk

no shutdown

interface fa0/5

switchport mode trunk

no shutdown

SW2 yapılandırma:

vlan 10

name LAN10

vlan 20

name LAN20

vlan 99

name Management

interface vlan 99

ip address 192.168.99.3 255.255.255.0

no shutdown

exit

ip default-gateway 192.168.99.1

interface fa0/18

switchport mode access

switchport access vlan 20

no shutdown

```
interface fa0/1  
switchport mode trunk  
no shutdown
```

R1 yapılandırma:

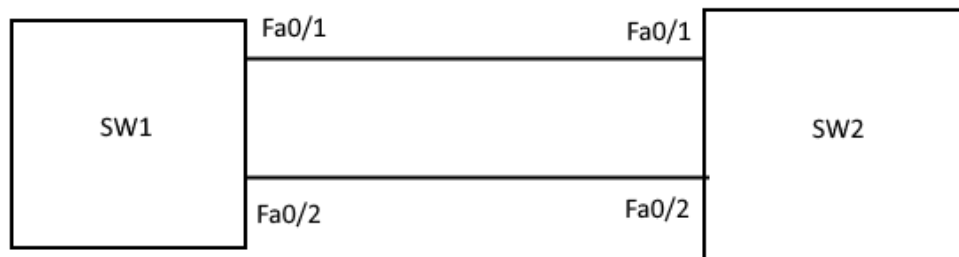
```
interface G0/0/1.10  
encapsulation dot1Q 10  
ip address 192.168.10.1 255.255.255.0
```

```
interface G0/0/1.20  
encapsulation dot1Q 20  
ip address 192.168.20.1 255.255.255.0
```

```
interface G0/0/1.99  
encapsulation dot1Q 99  
ip address 192.168.99.1 255.255.255.0
```

```
interface G0/0/0  
no shutdown
```

EtherChannel yapılandırma:



S1 için: (aynısı S2 için de yapılır)

```
interface range FastEthernet 0/1 - 2  
channel-group 1 mode active  
exit
```

```
interface port-channel 1  
switchport mode trunk  
switchport trunk allowed vlan 1,2,20
```

channel-group 1 mode DURUM komutunda

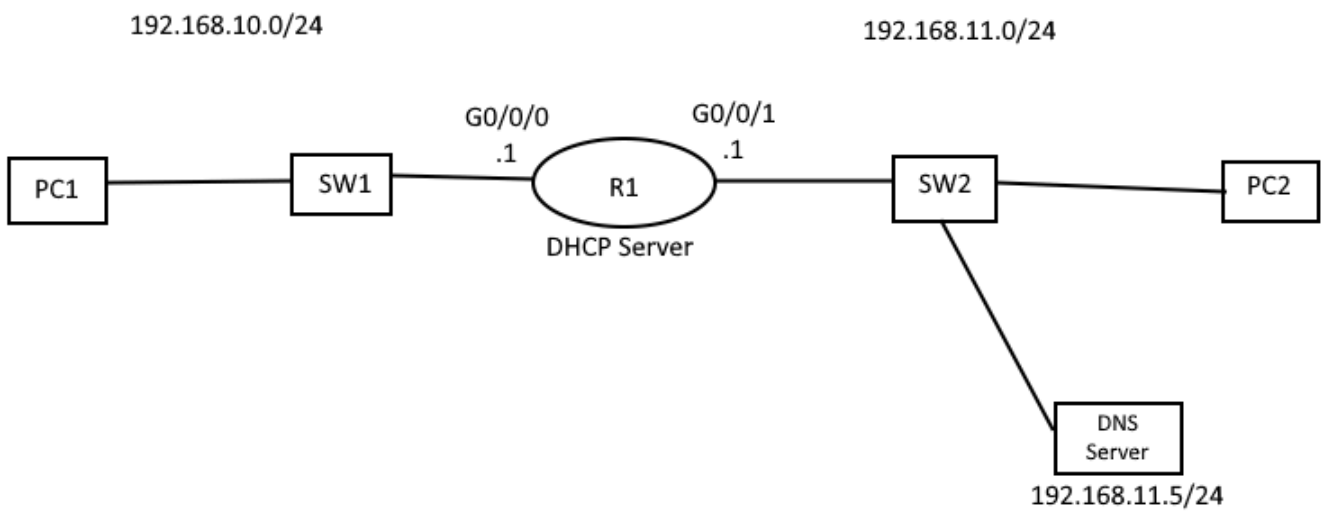
DURUM = active => LACP aktif

DURUM = passive => başka aygıttan LACP gelirse LACP aktifleşir

DURUM = desirable => PAgP aktifleşir

DURUM = auto => başka aygıttan PAgP gelirse PAgP aktifleşir

DHCP Sunucusu yapılandırma:



R1'in 192.168.10.0/24 LAN'ı için:

ip dhcp excluded-address 192.168.10.1 192.168.10.9 → 1 - 9 arası hariç

ip dhcp excluded-address 192.168.10.254

ip dhcp pool LAN-POOL-1

network 192.168.10.0 255.255.255.0

default router 192.168.10.1

dns-server 192.168.11.5

domain-name example.com

end

SHOW KOMUTLARI:

show ip route → Alt arayüzlerin yönlendirme tablosunda görünmesi
show ip interface brief → Alt arayüzleri IPv4 adresi ve up/down bilgisi
show interface ID → ID arayüzünün bilgilerini gösterir
show interfaces trunk → Switchteki etkin trunk portlarını gösterir

show vlan → VLAN'ları listeler
show vlan brief → VLAN adı, durumu ve port başına VLAN'ı görüntüler
show vlan ID → ID VLAN hakkında bilgi verir
show vlan summary → VLAN özet bilgilerini gösterir

show interface f0/18 switchport → f0/18 portunun veri/ses VLAN'ları doğru atanıp atanmadığını kontrol eder

show interfaces port-channel 1 → 1. port kanalı arayüzünün genel durumu
show etherchannel summary → port kanalı başına satır bilgi
show etherchannel port-channel → belirli bir port kanalı arayüzü
show interfaces f0/1 etherchannel → arayüzün rolü

show running-config section dhcp
show ip dhcp binding → Tüm IP'leri MAC'leri
show ip dhcp server statistics → Gönderilen ve alınan DHCPv4 mesaj sayısı

VLAN SALDIRILARINI AZALTMA

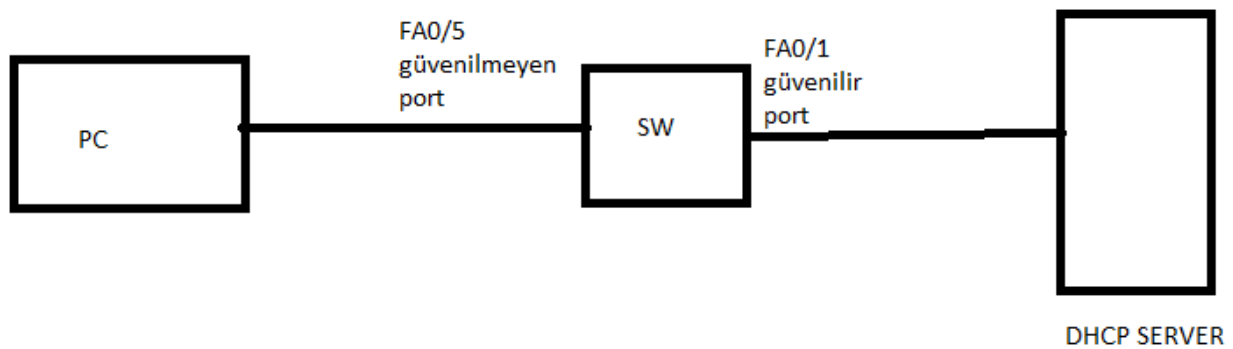
FA0/1 – FA0/16 Erişim portları
FA0/17 – FA0/24 Kullanılmayan portlar
FA0/21 – FA0/24 Trunk portlar
olmak üzere;

interface range fa 0/1 – 16
switchport mode access

interface range fa 0/17 - 20
switchport mode access
switchport access vlan 1000 (1000 = kullanılmayan bir VLAN)
shutdown

interface range fa 0/21 - 21
switchport mode trunk
switchport nonegotiate
switchport trunk native vlan 999

DHCP SALDIRILARI



SI'de:

ip dhcp snooping

interface fa 0/12

ip dhcp snooping trust

exit

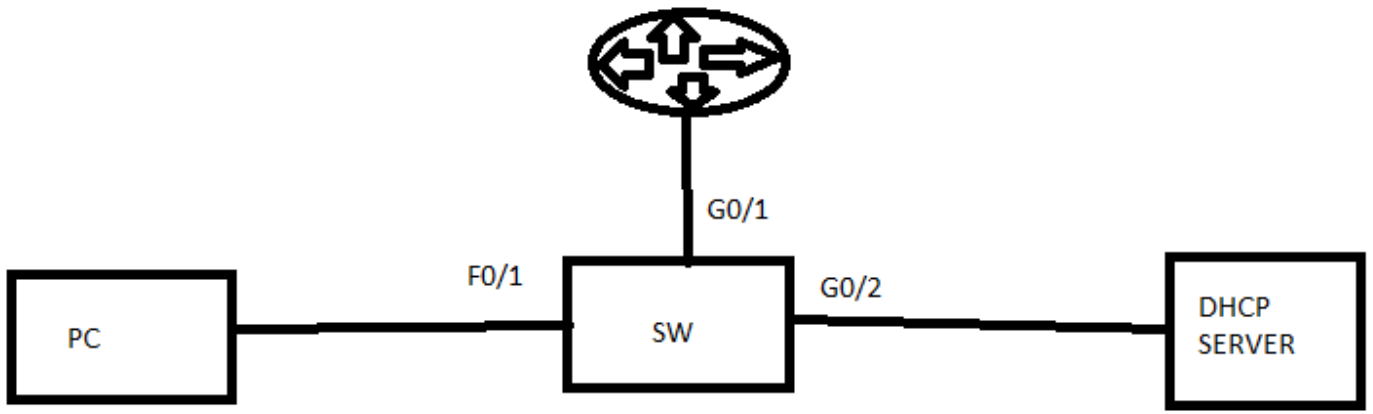
interface range f0/5 - 24

ip dhcp snooping limit rate 6

exit

ip dhcp snooping vlan 5,10,50-52 (5, 10, 50, 51, 52 VLAN'larda DHCP gözetleme aktif)

ARP SALDIRILARI



SW'de:

ip dhcp snooping

interface range g 0/1 - 2

ip dhcp snooping trust

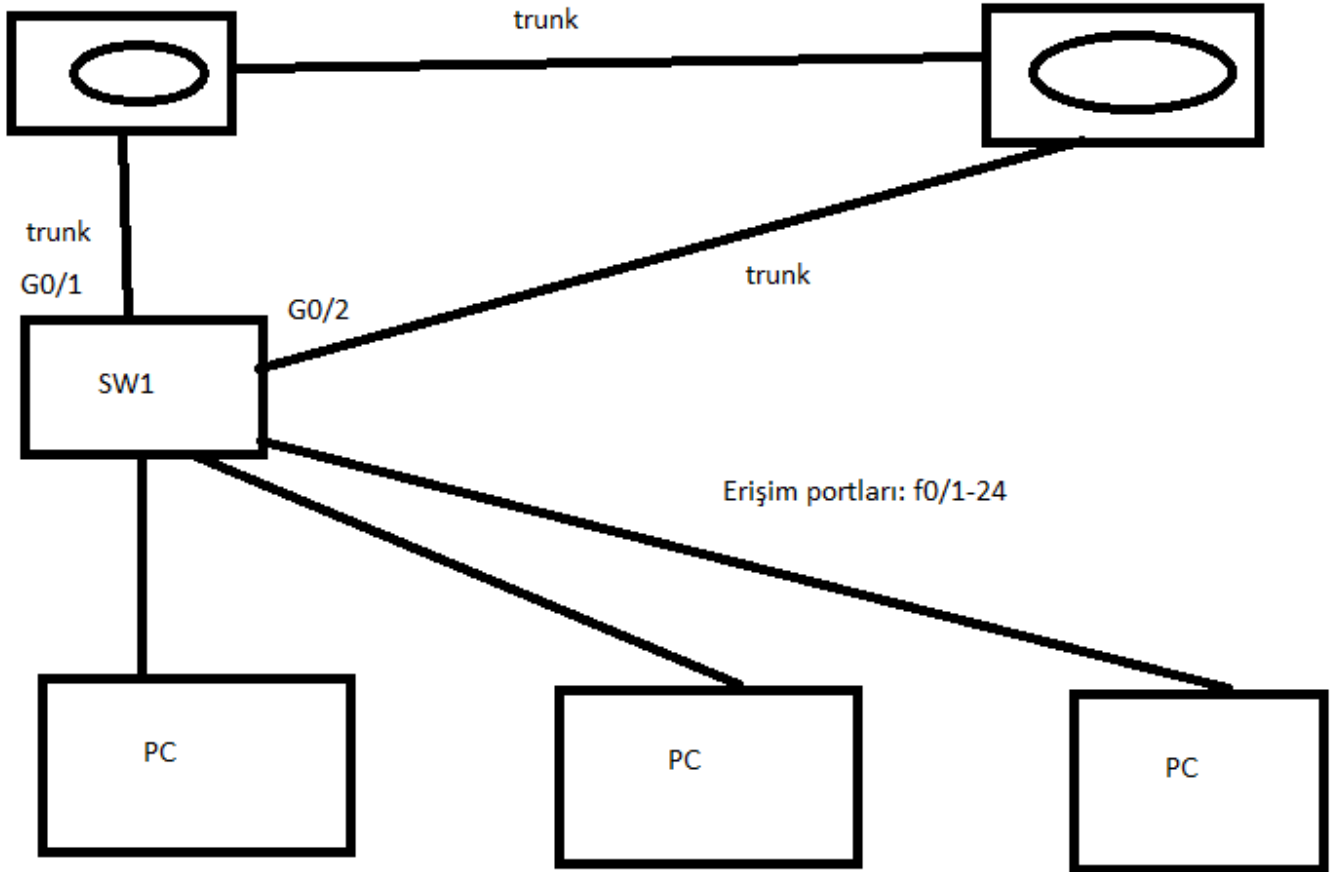
ip arp inspection trust

exit

ip dhcp snooping vlan 10,20,30-49

ip arp inspection vlan 10,20,30-49

STP SALDIRILARINI AZALTMA (son cihaza bağlı portlarda)



SW1'de:

```
interface range fa 0/1 - 24
```

```
switchport mode access
```

```
exit
```

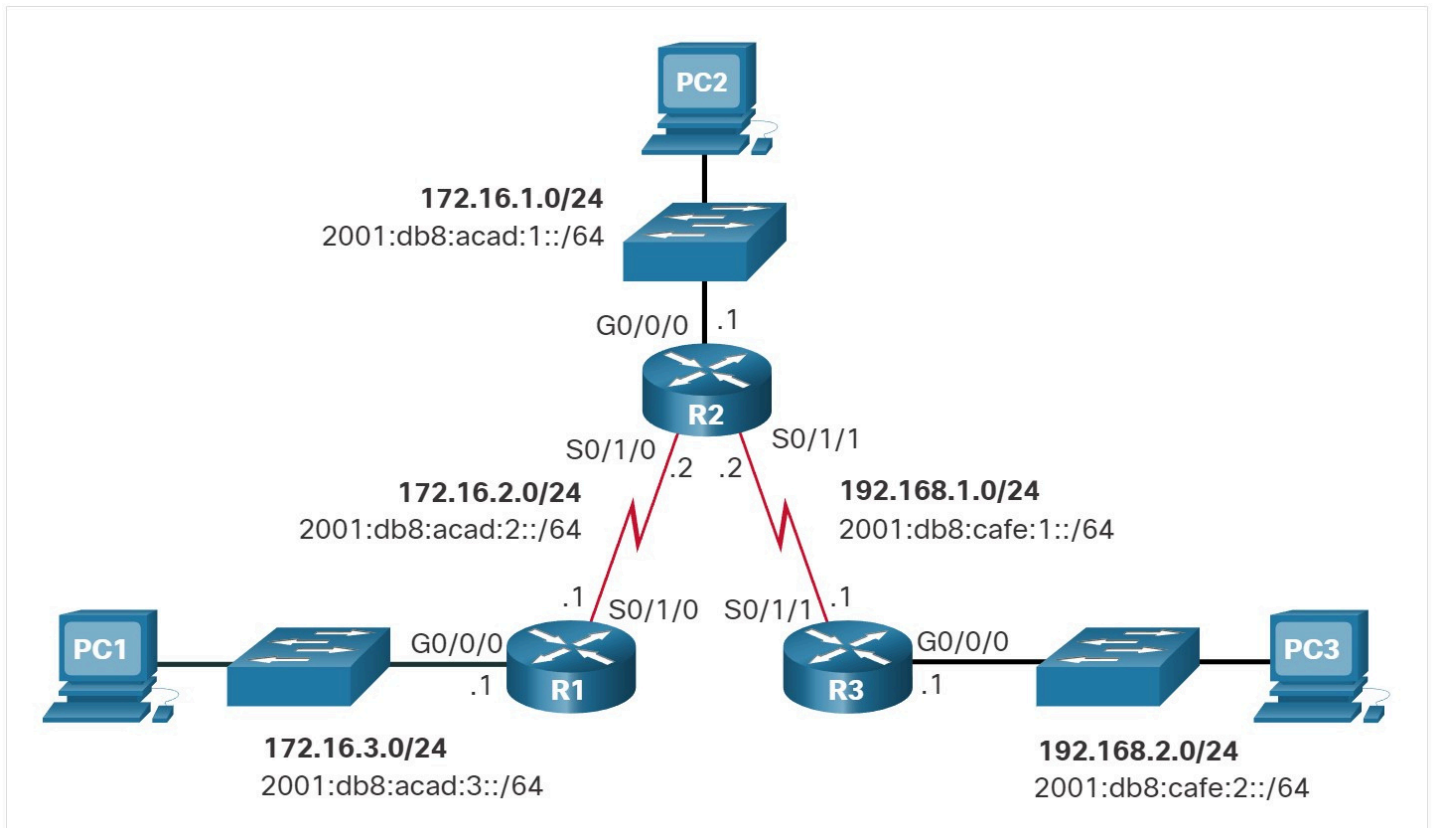
spanning-tree portfast default (trunk olmayan arayüzlerde aktif)

spanning-tree portfast bpduguard default (portfast aktif olan portlarda BPDU Guard aktif)

```
interface fa 0/1
```

```
spanning-tree portfast (sadece fa0/1 arayüzünde aktif)
```

IP STATİK YONLENDIRME



SONRAKİ SEKME BİÇİMİNDE AYARLAMA Rİ'DE:

```
ip route 172.16.1.0 255.255.255.0 172.16.2.2
ip route 192.168.1.0 255.255.255.0 172.16.2.2
ip route 192.168.2.0 255.255.255.0 172.16.2.2
```

ipv6 unicast-routing

```
ipv6 route 2001:db8:acad:1::/64 2001:db8:acad:2::2
ipv6 route 2001:db8:cafe:1::/64 2001:db8:acad:2::2
ipv6 route 2001:db8:cafe:2::/64 2001:db8:acad:2::2
```

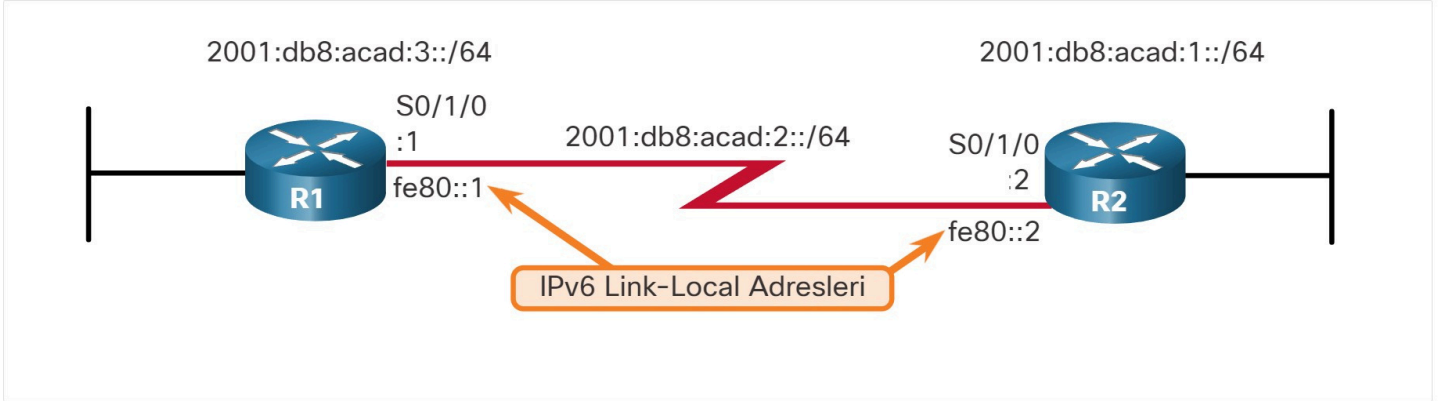
DOĞRUDAN BAĞLI BİÇİMİNDE AYARLAMA Rİ'DE:

```
ip route 172.16.1.0 255.255.255.0 s0/1/0
ip route 192.168.1.0 255.255.255.0 s0/1/0
ip route 192.168.2.0 255.255.255.0 s0/1/0
```

```
ipv6 route 2001:db8:acad:1::/64 s0/1/0
ipv6 route 2001:db8:cafe:1::/64 s0/1/0
ipv6 route 2001:db8:cafe:2::/64 s0/1/0
```

TAM BELİRGİN BİÇİMDE YAPILANDIRMA R1'DE:

```
ip route 172.16.1.0 255.255.255.0 GigabitEthernet 0/0/1 172.16.2.2
ip route 192.168.1.0 255.255.255.0 GigabitEthernet 0/0/1 172.16.2.2
ip route 192.168.2.0 255.255.255.0 GigabitEthernet 0/0/1 172.16.2.2
```



Tam belirgin IPv6 statik rotasında, hem çıkış arayüzü hem de sonraki sekme IPv6 adresi belirtilir.

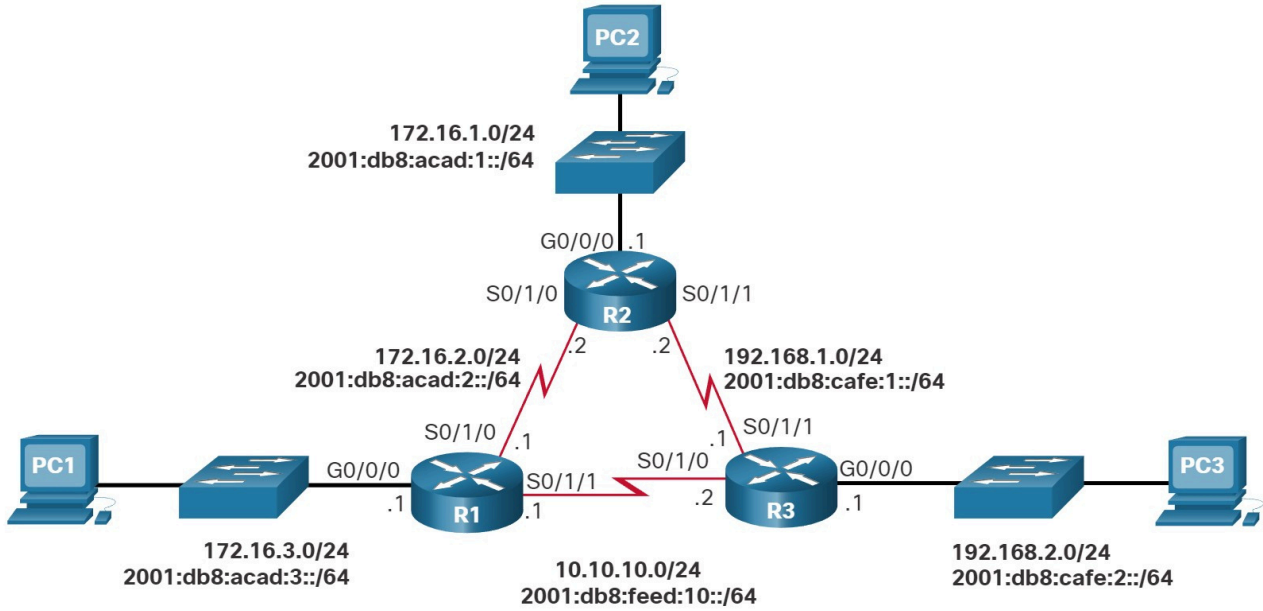
```
R1(config)# ipv6 route 2001:db8:acad:1::/64 s0/1/0 fe80::2
```

Varsayılan statik rota yapılandırma:

Daha belirgin rota girişleriyle eşleşmeyen tüm paketler, 172.16.2.2 adresli R2'ye iletilir.

```
ip route 0.0.0.0 0.0.0.0 172.16.2.2
ipv6 route ::/0 2001:db8:acad:2::2
```

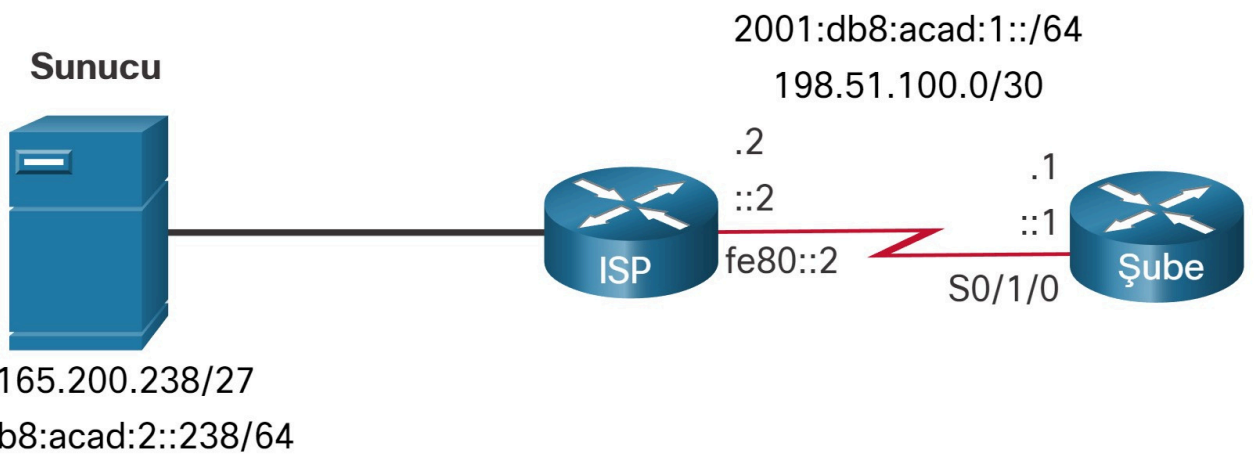
KAYAN STATİK ROTALAR:



```
R1(config)# ip route 0.0.0.0 0.0.0.0 172.16.2.2
R1(config)# ip route 0.0.0.0 0.0.0.0 10.10.10.2 5

R1(config)# ipv6 route ::/0 2001:db8:acad:2::2
R1(config)# ipv6 route ::/0 2001:db8:feed:10::2 5
```

STATİK HOST ROTASI YAPILANDIRMA:



```
şube(config)# ip route 209.165.200.238 255.255.255.255 198.51.100.2
şube(config)# ipv6 route 2001:db8:acad:2::238/128 2001:db8:acad:1::2
```